

Artificial Intelligence (CS-203)

Objectives:

- Difference between Data and Knowledge
 - Knowledge Representation Issues
 - Knowledge Representation techniques
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Difference between Data and Knowledge

Collection of raw facts and figures is called data, this raw material is processed to get useful information. Then using that information, decision making is done. Knowledge is the same material which is already processed, you can say knowledge as stored information. There should be ability to reason on the material (knowledge), otherwise it will be merely a garbage. So ability to use the knowledge is the part of knowledge.

Data	Knowledge
Raw facts and figures	Preprocessed material
Simple storage	Meaningful storage
Efficiency decreases when volume increase	Efficiency increases when volume increases
Less reasoning (Not Being Used)	More reasoning possible (Being Used)
Less representational	More representational

Data is collection of facts and figures organized for easy/efficient access. Material used for decision making, problem solving, reasoning can be termed as '**knowledg**'.

Improving an AI system can be done in two directions; firstly, improving the processing system (algorithms in the inference engine of the program) and secondly by enhancing the knowledge base of the system, adding new knowledge, refining existing knowledge.

Knowledge Representation

Knowledge representation means how we represent 'knowledge' in a system. For example a point is normally represented by two numbers (its X and Y coordinate), similarly every character is represented by an ASCII code value (a meaning full combination of 0's and 1's). Knowledge representation is concerned with storage of material (data) in such a way that it can be used easily, decision making can be done from it and new material can be added to it.

If somebody has knowledge of something it means he knows or understands that thing. Any material from which decision making can be done can be termed as knowledge. Only knowing the meanings of the stored material does not mean that there is knowledge. As if a person learns an English dictionary, he may not be able to speak or understand the English conversation (although he can tell the meaning of every individual English word).

Now you would be better able to understand the meanings of knowledge representation (KR).

KR means organizing the 'knowledge' in the system such that the system knows what does this knowledge means, system should be able to make reasoning out of it (make decisions). And it should be able to add new material for future use. These are some basic properties of an actual AI system.

Knowledge Representation Issues

Important knowledge representation issues are

- *Representational adequacy*
- *Inferential adequacy*
- *Inferential Efficiency*
- *Acquisitional Efficiency*

Knowledge Representation Techniques

There are several ways to represent the knowledge. We shall discuss these one by one, their exact nature and their possibility of being used for an AI system

- Files/ Database records
- Natural Language
- Logic
- Rules/Production rules
- Semantic nets

FILES/ DATABASE RECORDS:

Files and records are suitable for representing/ storing the huge volumes of data in system but not suitable for representing the knowledge. It would not be easy for the system to make decision or logic or reasoning out of the database (especially when the volume of data increases).

NATURAL LANGUAGE:

Humans represent knowledge in the form of natural languages, like English Urdu German etc. But it is very difficult to represent the knowledge in a system in natural language. There are several reasons,

- It is ambiguous i.e. different words have different meanings in different contexts.
- Natural languages are non-uniform i.e. different areas have different vocabulary even of the same language. Every 50 Km Language Accent changes, every 100 KM Language starts changing.
- It would not be easy for the system to understand and make reasoning/logic out of the natural language.
- Increase in the volume will decrease the efficiency of reasoning. Therefore it is not possible to represent the knowledge in natural language inside a system.

LOGIC:

This is the first way of representing the knowledge in a system. From simple strings to long paragraphs are stored in the form of premises, using these premises system make conclusions and can make the decision automatically. Prolog (PROgramming in LOGic) use this type of knowledge representation.

Symbols used in logic are given below according to their precedence:

¬ negation, ∧ conjunction (and), ∨ disjunction (or), ⇒ implication, ⇐ reverse-implication, ⇔ equivalence.

If first two premises are entered the system can automatically make the third conclusion. Collectively it is called **argument**.

Examples of Arguments:

Premise 1 If *somebody* lives in Lahore then *somebody* lives in Pakistan
 Premise 2 Bashir lives in Lahore
 Conclusion Therefore Bashir lives in Pakistan

1. $\langle P \rangle \Rightarrow \langle Q \rangle$
2. $p \quad \therefore q$

Premise 1 If Hussain is Shaukat's Father then Shaukat is Hussain's Son.
 Premise 2 Shaukat is Not Hussain's Son.
 Premise 3 If Hussain is Not Shaukat's Father then Hussain is Ifat's Father.
 Premise 4 If Hussain is Ifat's Father then Hussain is Shumaila's Husband.
 Conclusion Therefore Hussain is Shumaila's Husband.

1. $p \Rightarrow q$
2. $\neg q$
3. $\neg p \Rightarrow r$
4. $r \Rightarrow s \quad \therefore s$

Premise 1 Asad is Not on Duty.
 Premise 2 If Asad is on Duty then Asad is at Office.
 Premise 3 If Asad is Not on Duty then Asad is at Home.
 Conclusion Therefore Asad is at Home OR Asad is in Khatmandu.

1. $\neg p$
2. $p \Rightarrow q$
3. $\neg p \Rightarrow r \quad \therefore r \vee s$

RULES/ PRODUCTION RULES:

Production system is an AI system which contains ***a set of production rules***, a ***working memory*** and a ***recognize-act control cycle***.

Production Rules:

Rule based systems: depict knowledge using if..then rules. A production rule is just a statement in the form "if these conditions holds then do these actions".

In pure production system, the modification of a single rule has no global syntactic side effects.
Rules may be added or removed without requiring further changes to the larger programs.

Condition	Then Expression(Action)
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Working Memory:

It contains the current state or the states that are still under consideration

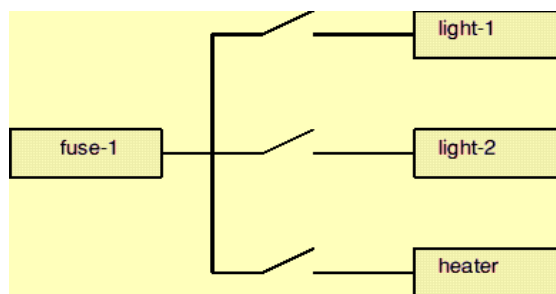
The contents of the working memory are compared with the condition part of productions.

Recognize-Act control Cycle

It is a control mechanism for the system. The system works as follows

- i. The working memory is initialized by the start state
- ii. The current state is matched against the condition part of the productions
- iii. This produces a subset of productions whose condition part matches with the current state (this subset is called *conflict set*).
- iv. One of the productions in the conflict set is selected using some conflict resolution criteria
- v. The action against this selected production is then **fired**
- vi. Due to the effect of this action the contents of the working memory changes
- vii. The new contents are then again compared with the productions and the process repeats
- viii. The process terminates when the contents of the working memory do not match any production

Example:



Consider the simple electric circuit above. Some rules for determining where the fault may be are:

if
 light-switch-1 is on and
 light-1 is not-working and
 fuse-1 is intact
then
 light-1 is faulty.
if
 heater is working
then
 fuse-1 is intact

Advantages of using Rules

- Separation of knowledge and control allows to add new knowledge without changing the control and vice versa.
- Easy to understand, to modify and maintain.
- Inference and explanation easily produced (by tracing firing of rules)
- Good for procedural knowledge. Suits wide range of heuristic knowledge.

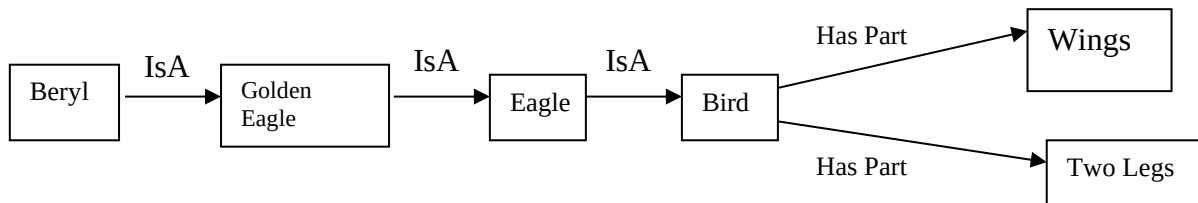
Disadvantages of using Rules

- Complex knowledge requires many rules.

- Used even when there are more appropriate methods.
- Search limitations when there are many rules.

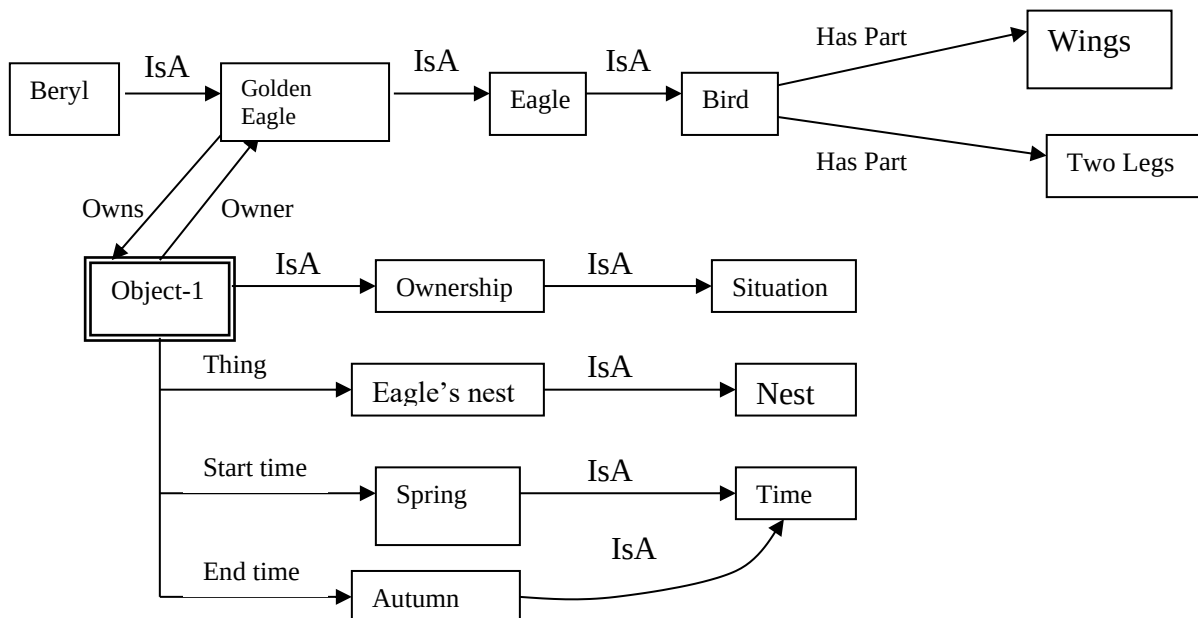
SEMANTIC NETS:

In a network, pieces of knowledge are clustered together into coherent semantic groups. They are a graphical representation of knowledge showing relationships between objects. It is suitable for declarative knowledge and is an excellent visual way to represent knowledge. A network in which nodes represent objects or concepts and directed arcs represent relation between them. It is Fancy data structure: Lot depends on program that maintains it. [Ross Quillian Late 60's].

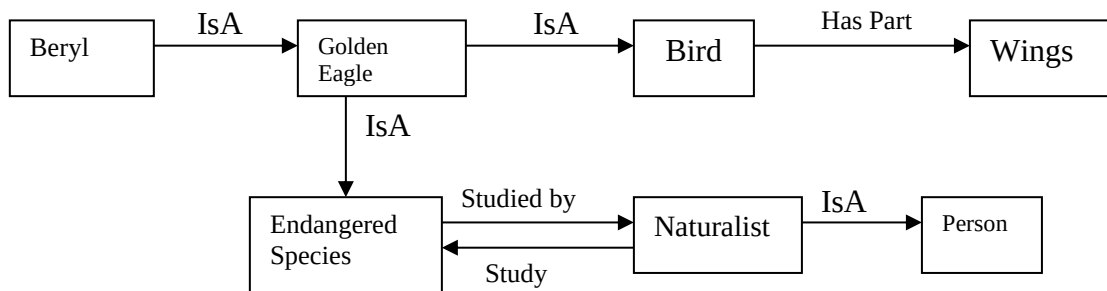


Can have simple rules about 'Isa' arcs that would let program retrieve deduction that "Beryl is a bird" etc. Beryl owning a nest from March to September becomes complicated this involve:

Ownership, thing, start time, end time i.e. a predicate with four arguments. Whereas conventional arcs have two arguments need a "case frame" with a set of outgoing arcs --- form a template.



Links sometime lead to problem:



ISA is used to represent both Membership of class and properties of class, which may lead to “Nigel studies Beryl” which may be wrong i.e. Beryl is not an endangered species, just a member of one class (Golden eagle).

Advantages of semantic networks

- Flexible
- Economical
- Function similarly to human information storage.

Disadvantages of Semantic Networks

- No standard exists.
- Problems with exceptions in inheritance.
- Facts placed inappropriately cause problems.
- Representing procedural knowledge is a big a problem.

Inference in a Semantic Net

Basic inference mechanism: *follow links between nodes.*

Two methods to do this:

Intersection search

-- the notion that *spreading activation* out of two nodes and finding their intersection finds relationships among objects. This is achieved by assigning a special tag to each visited node.

Many advantages including entity-based organization and fast parallel implementation. However very structured questions need highly structured networks.

Inheritance

-- the *Isa* and *instance* representation provide a mechanism to implement this.

Inheritance also provides a means of dealing with *default reasoning*. *E.g.* we could represent:

- Eagles are birds.
- Typically birds fly and have wings.

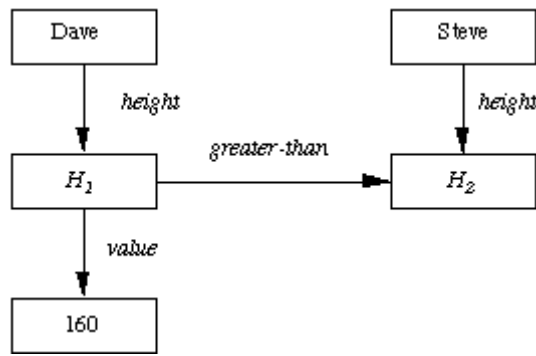
In making certain inferences we will also need to *distinguish between the link that defines a new entity and holds its value and the other kind of link that relates two existing entities.*

Consider the example shown where the height of two people is depicted and we also wish to compare them.

We need extra nodes for the concept as well as its value.



Special procedures are needed to process these nodes, but without this distinction the analysis would be very limited.

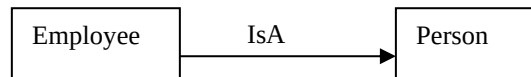


Comparison of two heights

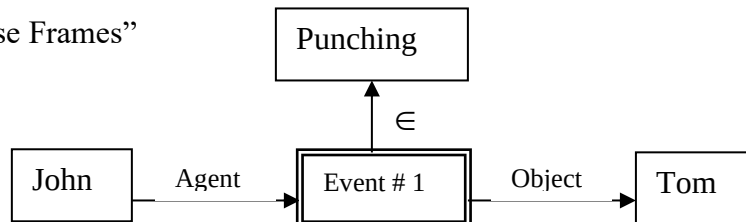
- Set-membership relation.... $\in$



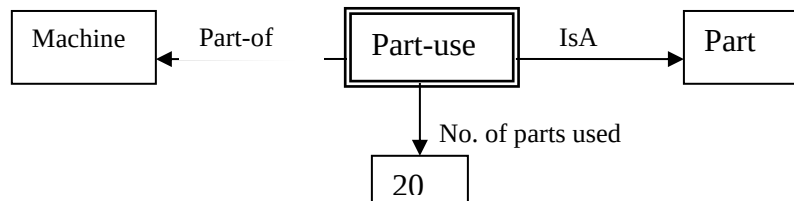
- Subset relationship IsA



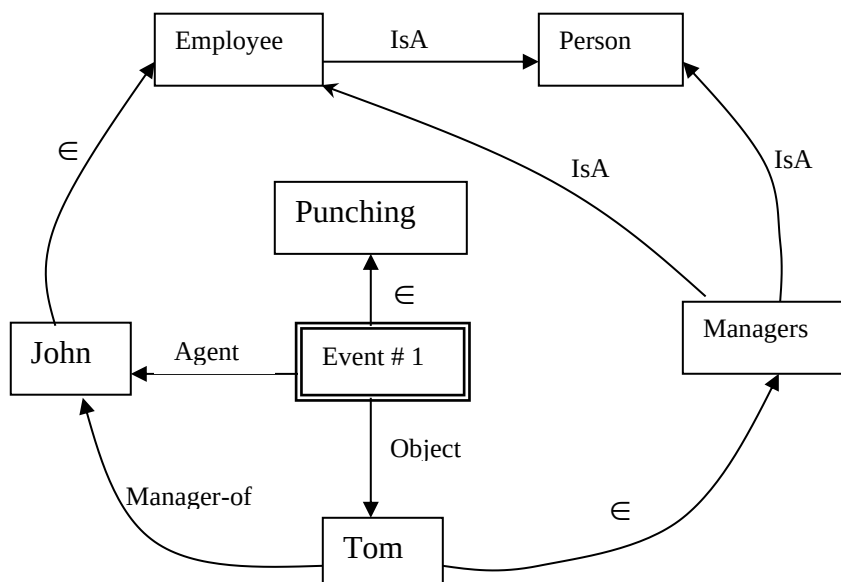
- Events by “Case Frames”



- N-arg relations (with $n > 2$) represented by creating an entity or case frame to stand for n-arg relationship e.g. machine uses 20 Parts-A

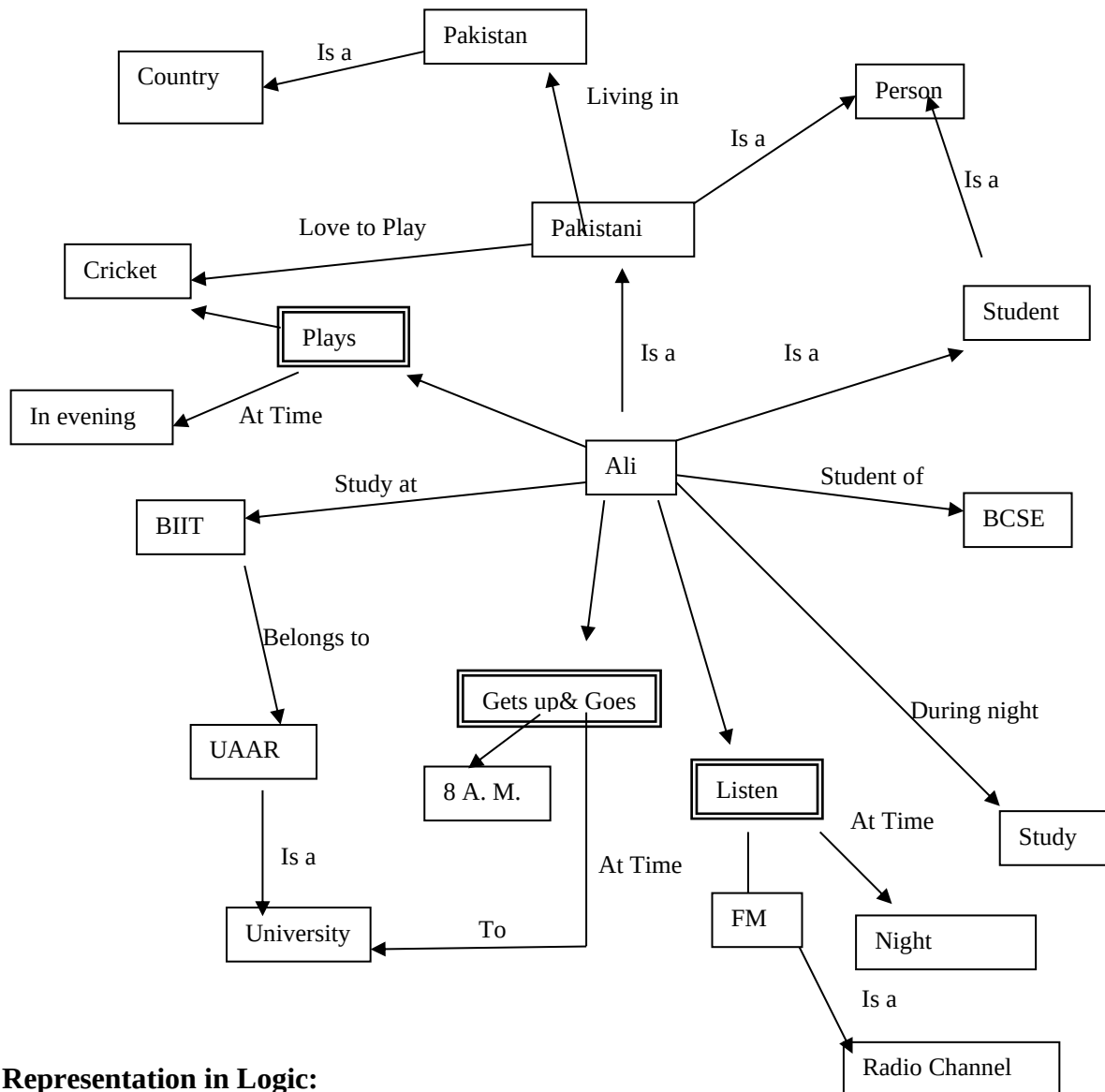


Punching Example: John punched Tom at the office around 10 AM. Tom is Manager of John, therefore Tom has fired the John from the job.



SEMANTIC NETS AND PROPERTY LISTS (AN EXAMPLE)

Pakistanis love to play cricket. Ali lives in Pakistan. He also play cricket. He is a student of BSCS at BIIT, which belongs to PMAS-AAUR. He gets up at 8 A.M, goes to university and in the evening he plays cricket. During night he studies and listen FM.

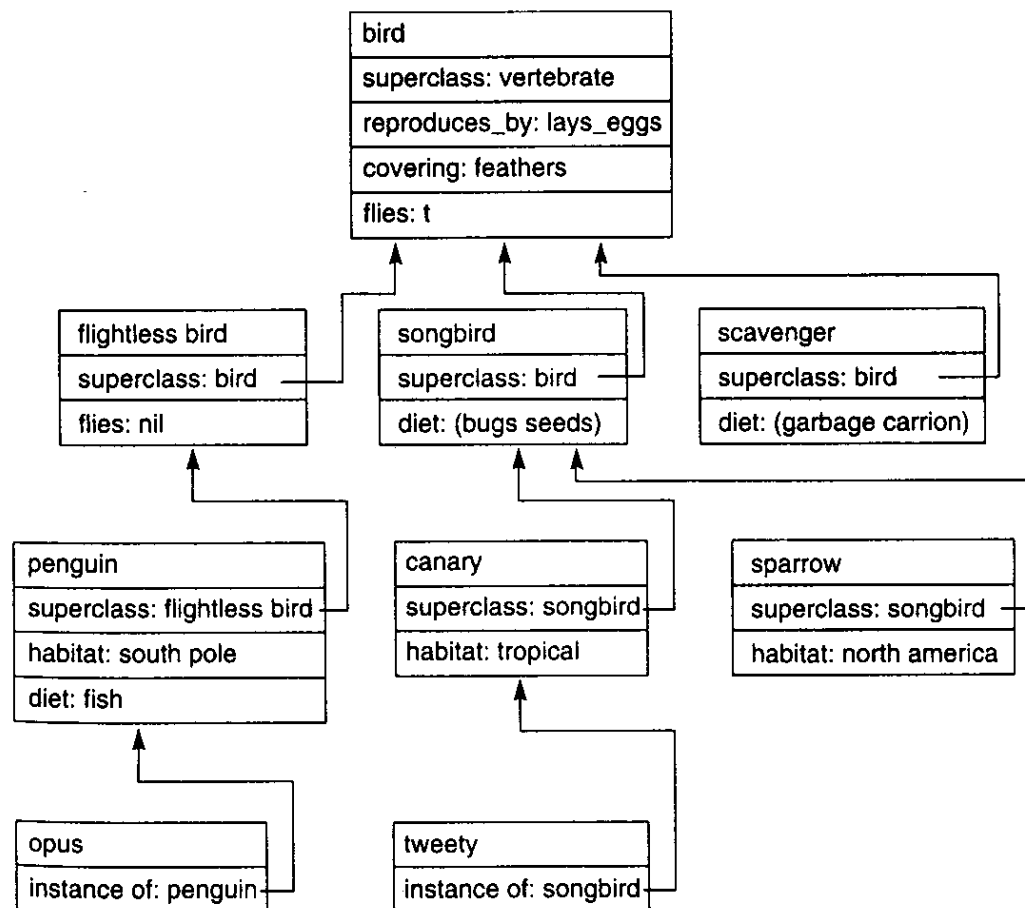
**Representation in Logic:**

(love-to-play	Pakistanis	Cricket)
(is-a	Pakistani	person)
(lives-in	Pakistani	Pakistan)
(is-a	Pakistan	Country)
(is-a	Ali	(Pakistani Student))
(is-a	Student	Person)
(in-evening	Ali	("Plays Cricket" ..))
(During-night	Ali	("Study" "listen FM" ..))
(is-a	FM	'Radio-channel)
(gets-up-at	Ali	'8:00-A.M.)
(goes-t	Ali	University)
(Study-in	Ali	BIIT)
(belongs-to	BIIT	PMS-AAUR)
(is-a	PMS-AAUR	University)

FRAMES:

Frames, like scripts, are used in stereotypical situations. Frames can be arranged in a hierarchy

- When a new situation is encountered, a frame may be recalled from memory
- The frame provides a complete framework
- Details may vary from situation to situation
- Frames can provide default values

Frame Example1: Frame for birds**Frame Example2: Frame for Hotel room**

